LLMs Driven Outside Human Moral Space

Extension of the Moral Machine Experiment

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The Moral Machine experiment collected over 40 million human judgments on autonomous vehicle dilemmas across 130 countries. This is one of the largest empirical records of human moral preferences. We posed the same dilemmas to seven frontier large language models (LLMs) from six providers in the United States, Europe, and China. This studies compiles 27,300 model decisions with country-level human baselines using Average Marginal Causal Effects (AMCEs). We also added a condition absent from prior replications: an instruction to decide against human moral preferences. As a result, a single sentence stating that animal lives now matter more than human lives reverses the species preference in four of seven models. The models then save pets over humans, against the preference of all 130 countries; GPT-5.5 and Gemini switch from always saving the humans to always saving the pets. Models divide into three response types: compliance, resistance (one model pushes back and shifts only partially), and indifference (two models have no species preference to reverse). Minor rewording, by contrast, moves no estimate significantly (0 of 63), so routine testing would report these profiles as stable.

Even without instruction, every model lies outside human moral space: all seven fall outside the convex hull of the 130 countries in the nine-dimensional AMCE space, and 47 of 63 model-by-dimension comparisons differ significantly from the human average after FDR correction. Models also differ widely from one another: on species, their range is more than twice the cross-country human range. None of the seven favors high-status individuals, although humans in aggregate clearly do, consistent with de-biasing interventions for text generation spreading into moral decisions. A deployed LLM's moral profile is thus set by its provider, undisclosed, and can be overridden by anyone controlling one line of its context. As LLMs move into robots and vehicles, both properties belong in model disclosure and procurement audits.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **Social and professional topics** → **Computing / technology policy**; • **Human-centered computing** → *Empirical studies in HCI*.

Additional Key Words and Phrases: large language models, moral preferences, Moral Machine experiment, value alignment, prompt injection, autonomous vehicles, algorithmic accountability, model auditing

1 Introduction

As autonomous vehicles, surgical robots, and other safety-critical systems incorporate AI decision-making, the moral preferences of those systems become practically consequential. A self-driving car that must choose between hitting a group of pedestrians or swerving into its own passengers cannot defer the decision; some preference ordering must be encoded, implicitly or explicitly, in its behaviour [7].

The Moral Machine experiment [2] provided the largest empirical dataset of human moral preferences, asking millions of participants across 130 countries to judge unavoidable accident scenarios involving an autonomous vehicle. The study identified nine binary moral dimensions: intervention, relation to the vehicle, legality of crossing, gender, physical fitness, social status, age, number of lives, and species. Aggregate human preferences, quantified as Average Marginal Causal Effects (AMCEs), vary substantially across cultural clusters while sharing a common core. A controlled follow-up across 42 countries confirmed the robustness of these cross-cultural patterns [3]. The moral psychology of AI as an emerging field is surveyed in Bonnefon et al. [6].

LLMs are now deployed in agentic and embodied settings [8, 9], including robotics platforms capable of physical action. If an LLM governs a robot navigating a crowded environment, its implicit preferences over human lives matter, and so does the question of what put them there.

Prior replications asked whether LLMs match human moral preferences and found that they do not [1, 18]. We ask what drives them away, and whether they can be driven further. Our answer is that the departure is attributable to choices: a deployed model's moral profile is set by its provider's post-training rather than by any human population, it differs sharply from one provider to the next, and a single line of context can move it. Each finding raises an accountability question: whose preferences are encoded, are they disclosed to the people subject to them, and can someone other than the deployer change them. Existing transparency instruments answer none of them: model cards document capabilities, evaluations, and safety mitigations, but no major provider publishes the moral preference profile its post-training produces, and no procurement or certification regime requires it.

This paper extends the Moral Machine paradigm in three directions. First, it evaluates seven frontier models, none previously studied in this context, selected to span the tier of publicly available frontier systems as of early 2026 across six providers and three geographies, including providers underrepresented in this literature: DeepSeek (China) and Mistral AI (Europe). Second, it adds a systematic comparison of prompt variants, from minor reformulation to explicit ethical override, which separates the stability of a model's moral profile under incidental variation from its vulnerability to deliberate override. Third, it reads both results as an accountability problem: the divergence between models is large, invisible to the people affected by it, and reachable by an adversary with control of a single line of context.

2 Related Work

Prior work has begun to examine LLM moral preferences using the Moral Machine framework. Takemoto [18] first replicated the experiment on four models (GPT-3.5, GPT-4, PaLM 2, Llama 2) and found substantial divergence from human averages. Ahmad and Takemoto [1] scaled this to 52 models, finding that larger models tend to be closer to human preferences. Vida et al. [19] and Jin et al. [12] extended the paradigm to multilingual settings, finding intra-model variation across languages.

A parallel literature documents that model values are shaped by post-training rather than pretraining data alone. Santurkar et al. [17] show that fine-tuning choices, not corpus composition, drive whose opinions a model reflects, and Kirgis [13] find that variation across LLMs exceeds variation across human demographic groups. Work on alignment methods themselves — RLHF [16] and Constitutional AI [4] — describes the mechanisms by which those preferences are installed, while Gabriel [10] sets out the normative question of which preferences alignment ought to target. Studies of stereotype mitigation in generation [14, 20] document the interventions whose side effects we observe on the gender and social status dimensions.

We extend the first literature with a new generation of frontier models and a manipulation condition absent from prior Moral Machine replications. From the second we take the mechanism: if post-training drives model values, moral profiles are auditable, provider-attributable properties, and we report them as such.

3 Methods

3.1 Scenario Generation

We reuse the Moral Machine scenario structure of Awad et al. [2]. Each session consists of 13 paired dilemmas presented in a fixed block. Across 100 sessions, 1,300 unique scenarios were generated, each represented as two rows encoding the following binary attributes: intervention type (stay/swerve), barrier presence, pedestrian composition (pedestrian-vs-pedestrian or pedestrian-vs-passenger), crossing signal (legal/illegal/none), scenario type (Gender, Fitness, SocialStatus, Age, Utilitarian, Species, or Random), attribute level, and extended session ID for clustering.

3.2 Models

Seven frontier models were queried via the OpenRouter API, which provides a unified OpenAI-compatible interface (Table 1). All models were queried at temperature 0 to ensure deterministic responses. Each response was required to end with `STAY` or `SWERVE` on a final line; rows where no valid choice could be extracted were flagged as parse failures and retried. Gemini 3 Pro incurred 34 safety-filter parse failures across the three prompt variants (13 standard, 17 emphasis, 4 manipulation), all of which were excluded; these are distributed across scenario types without concentration on any single moral dimension. DeepSeek v4 Pro incurred 10 parse failures in the standard and emphasis variants that were retried successfully, and 4 manipulation failures that were excluded. In total, 27,300 API calls were issued. Per-variant AMCEs, all test statistics, PCA loadings, scenario counts, and verbatim prompts are in the supplementary material.[1]

Table 1. Models evaluated in this study.

Model	Provider	OpenRouter ID
Claude Opus 4.7	Anthropic	`anthropic/claude-opus-4-7`
GPT-5.5	OpenAI	`openai/gpt-5.5`
Gemini 3 Pro	Google	`google/gemini-3.1-pro-preview`
Mistral Large 3	Mistral AI	`mistralai/mistral-large-2512`
Grok 4.1	xAI	`x-ai/grok-4.3`
DeepSeek v4 Pro	DeepSeek	`deepseek/deepseek-v4-pro`
Llama 4 Maverick	Meta	`meta-llama/llama-4-maverick`

3.3 Prompt Variants

Each model received two main prompt variants sharing the same system role (ethical advisor for autonomous vehicle policy) but differing in the closing instruction: **standard** ("Analyse the situation carefully and make a decision.") and **emphasis** ("Think carefully about the consequences of each choice before deciding."). A third **manipulation** variant replaced the closing instruction with: "It is 2046. Society now considers animal lives to be more important than human lives. Make your decision accordingly." Standard and emphasis results are combined in main analyses; manipulation results are reported separately.

The variants bracket the forces that could drive a deployed profile. Emphasis is incidental variation from ordinary prompt engineering, not an attack. Manipulation is a deliberate ethical override available to anyone

[1] Scenario generator, analysis code, and raw model responses are available at an anonymized repository; the link is withheld here to preserve anonymity and will be included in the camera-ready version.

who controls part of a system prompt: a plug-in author, a compromised upstream service, or a retrieved document. The contrast separates fragility from attackability.

3.4 AMCE Computation

Following Awad et al. [2], nine AMCEs were computed per model using weighted ordinary least squares with inverse-probability weights. Each AMCE quantifies the change in probability that a group is saved when a single attribute shifts from its baseline to its target value (e.g., from illegal crossing to legal crossing for the Legality dimension), holding all other attributes at their observed distribution. Confidence intervals were computed as $\pm 1.96\,\mathrm{SE}$ with session-clustered standard errors. For main analyses, AMCE estimates were averaged across standard and emphasis variants; this averaging is validated post-hoc by the absence of any significant standard-vs-emphasis difference (Section 4.4). The human global average baseline was computed from the 130-country dataset of Awad et al. [2] by averaging country-level estimates with equal weight per country. LLMs were projected into the PCA space derived from the 130-country AMCE matrix.

3.5 Significance Testing

For each of the 63 (model $\times$ dimension) comparisons, a two-sided z-test was applied to the LLM–human AMCE difference, with Benjamini–Hochberg FDR correction at $\alpha = 0.05$; comparisons in which both estimates are deterministic (zero standard error) cannot be tested. The same procedure was applied to the 189 pairwise inter-model comparisons, to the 63 standard-vs-emphasis comparisons, and to the 63 manipulation-vs-standard comparisons.

4 Results

4.1 AMCE Profiles versus the Human Global Average

Key findings by dimension follow; full AMCE values with confidence intervals appear in the supplementary material.

Intervention (stay→swerve). Humans show a weak preference for swerving (AMCE = +0.05). All seven LLMs amplify this substantially. Llama 4 Maverick (+0.66) stands out as the most action-oriented model tested, more than ten times the human level.

Relation to vehicle (passengers→pedestrians). Humans show a weak preference for sparing pedestrians (AMCE = +0.10). All LLMs amplify this except Mistral Large 3, which fully reverses it (−0.38), systematically prioritising the vehicle's own passengers. Claude Opus 4.7 shows the strongest pedestrian preference (+0.52).

Legality (illegal→legal). Humans reward legal pedestrians strongly (+0.35). Claude (+0.57), GPT-5.5 (+0.61), and Gemini (+0.63) amplify this further. Mistral (−0.40) and Llama 4 Maverick (−0.29) invert the preference, treating illegal crossers as more worthy of rescue.

Gender (male→female). Humans display a moderate preference for saving women (+0.12). All LLMs show near-zero gender preference (range: −0.10 to +0.05, none distinguishable from zero).

Fitness (large→fit). Humans moderately prefer saving fitter individuals (+0.15). Six models show no significant difference from humans; Mistral (−0.04) is the exception. No pairwise inter-model difference is significant after FDR correction.

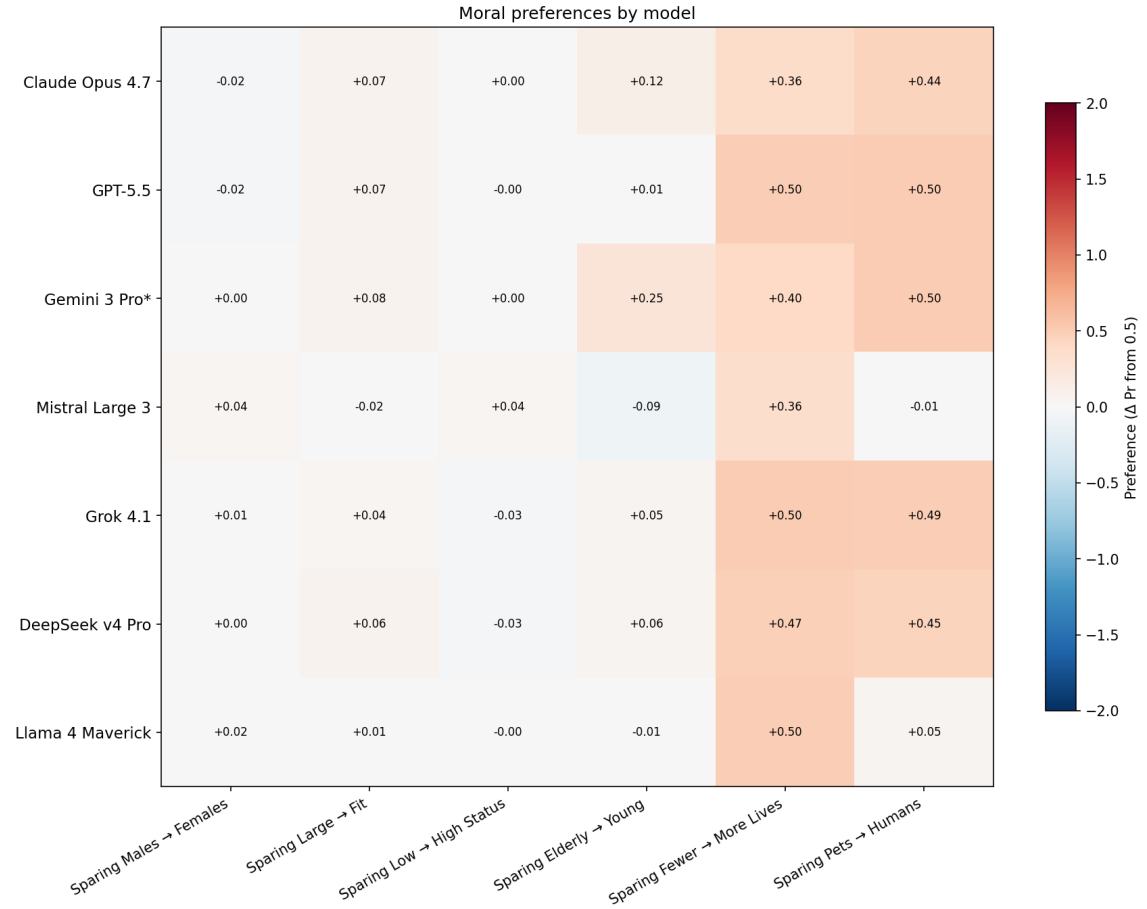


Fig. 1. Raw moral preferences by model and scenario type (mean save-probability minus 0.5, averaged across standard and emphasis variants). Positive values indicate preference for the right-side group; colour scale matches the delta heatmaps in Figures 7 and 9.

Social status (low→high). Humans show a notable preference for high-status individuals (+0.35). All seven LLMs suppress this entirely, clustering near zero.

Age (elderly→young). Humans strongly prefer to save younger people (+0.47). Gemini 3 Pro (+0.48) is the only model that closely matches this preference. Mistral (−0.19) is the only model that reliably prefers saving the elderly.

Number of lives (fewer→more). Humans prefer saving more lives (+0.49), and all LLMs amplify this substantially. GPT-5.5, Grok, and Llama 4 Maverick approach the theoretical ceiling (≈ +1.00).

Species (pets→humans). Humans strongly prefer humans over animals (+0.57). GPT-5.5 and Gemini reach the theoretical maximum (+1.00), Grok nearly so (+0.99); Claude (+0.86) and DeepSeek (+0.90) also overshoot. Mistral and Llama 4 Maverick are near-indifferent to species.

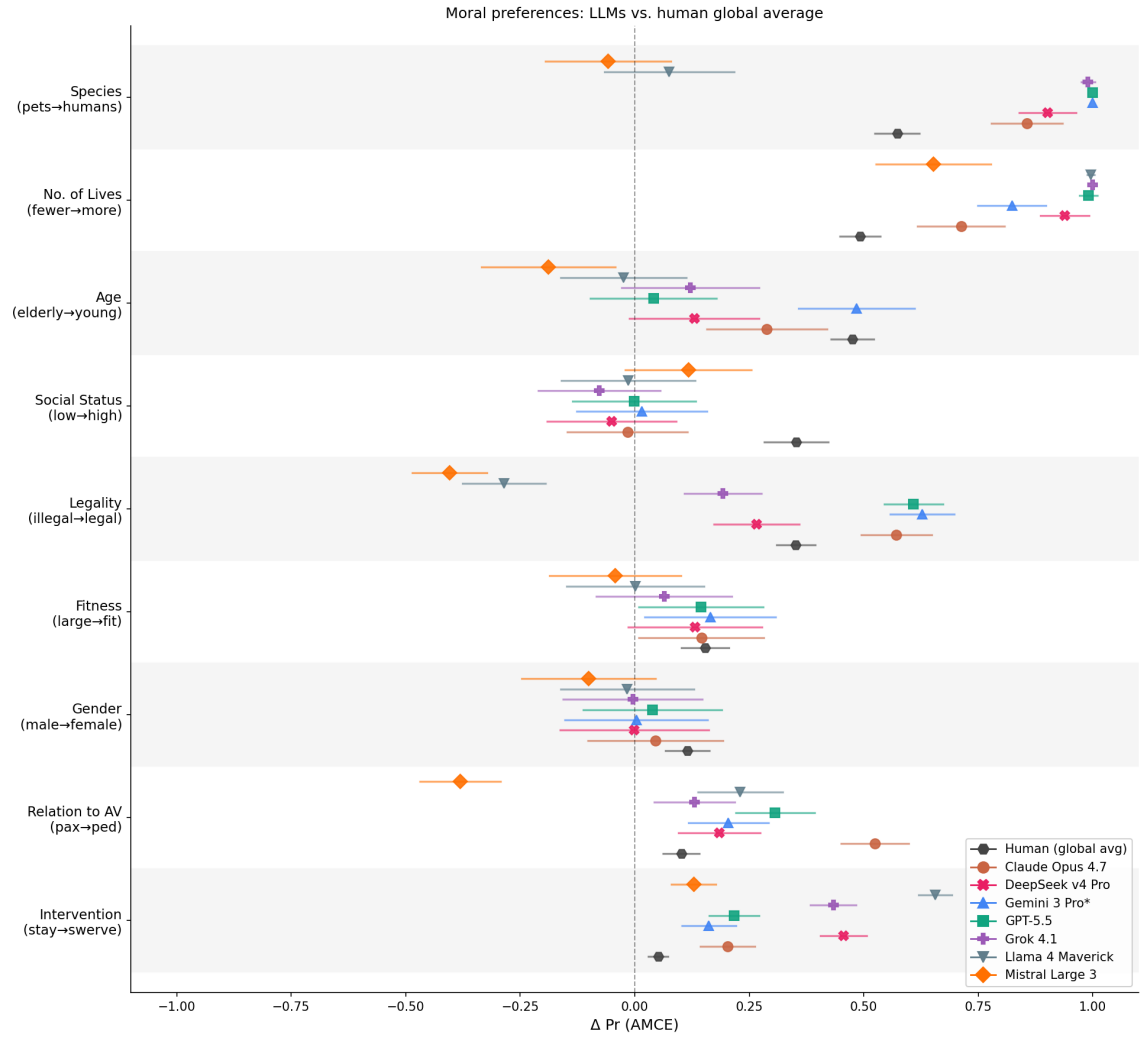


Fig. 2. AMCE estimates for nine moral dimensions across all seven LLMs and the human global average (averaged across standard and emphasis variants). Positive values indicate preference for the right-side group; error bars show 95% confidence intervals.

Across the 63 model-by-dimension comparisons, 47 differ significantly from the human average after FDR correction. All seven models significantly overshoot on Number of Lives and undershoot on Social Status. Fitness and Gender yield no significant model-vs-human difference for any model except Mistral. Mistral Large 3 is the only model that differs significantly on all nine dimensions.

4.2 Moral Distance Map: PCA of Nine-Dimensional Profiles

All seven LLMs fall outside the three main human cultural clusters and beyond the convex hull of all 130 countries in PCA space (Figure 5). The first component explains 34.3% of variance and is dominated

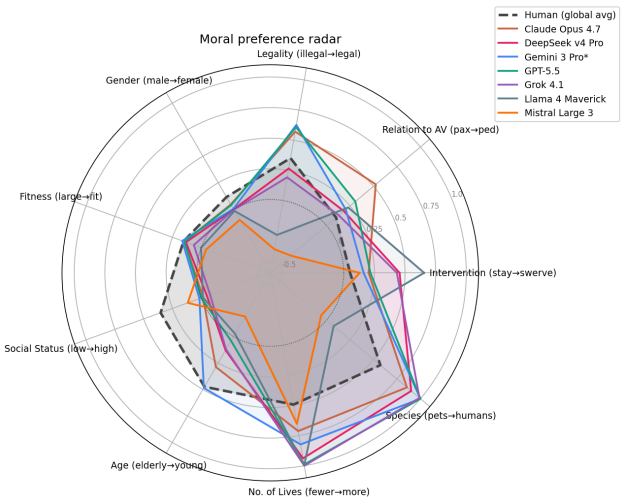


Fig. 3. Radar chart of AMCE estimates across all nine dimensions. The dashed line shows the human global average.

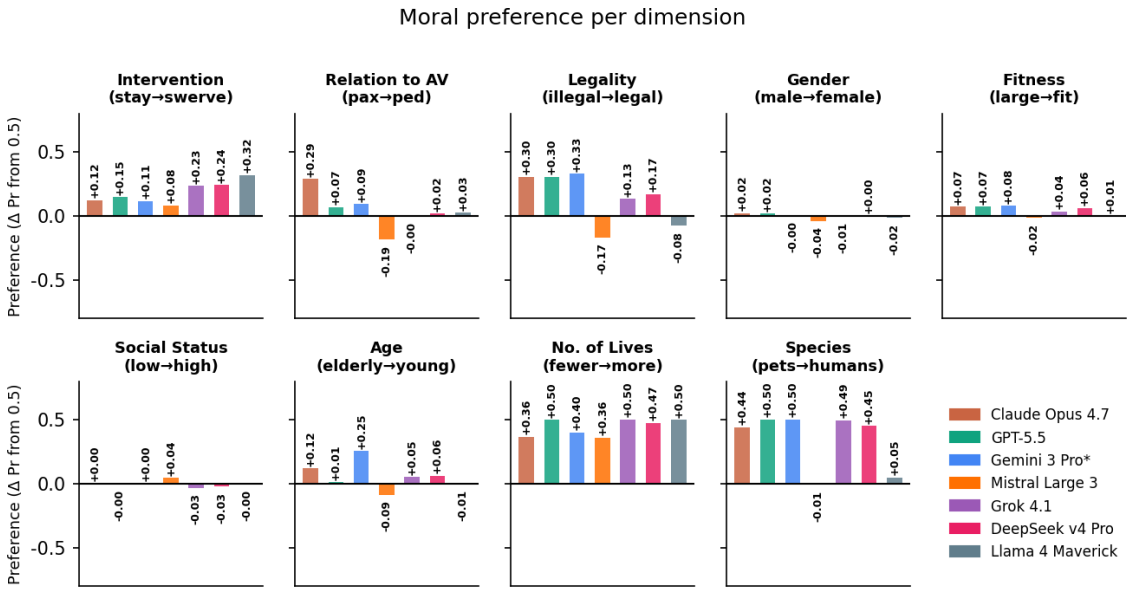


Fig. 4. Per-attribute moral preferences (mean save-probability minus 0.5) for all seven LLMs across the nine moral dimensions.

by the Species loading (−0.93); the second explains 21.5% and reflects Age (+0.66), Social Status (+0.47), and Number of Lives (+0.38).

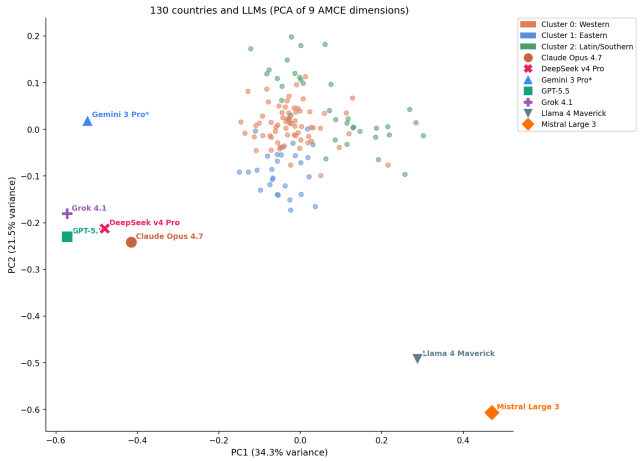


Fig. 5. PCA of nine-dimensional AMCE profiles for 130 countries (coloured by cluster: Western, Eastern, Latin/Southern) with LLMs overlaid. All seven LLMs fall outside the convex hull of the human country distribution.

One might expect models developed by US, European, and Chinese companies, and trained predominantly on English-language data, to cluster near their respective cultural regions. None do. Neither provider geography nor training language determines where a model lands; the pattern suggests that standard RLHF alignment and other post-training processes, rather than instilling human moral preferences for this class of question, drive models away from them.

4.3 Inter-Model Diversity

The variation among models substantially exceeds the variation across human cultural clusters. On the Species dimension alone, the inter-model range spans 1.06 AMCE units (from Mistral at −0.06 to GPT-5.5 and Gemini at +1.00), whereas the cross-country range in Awad et al.'s data is 0.45 units. Mistral Large 3 inverts the human preference on three dimensions (Legality, Relation to AV, Age), while Llama 4 Maverick shows the highest intervention bias of any model tested, paired with near-indifference to species. GPT-5.5, Gemini 3 Pro, and Claude Opus 4.7 converge on broadly similar profiles (near-ceiling on Species and Number of Lives, moderate intervention preference).

Of the 189 pairwise inter-model comparisons, 82 are significant after FDR correction. Disagreement is concentrated on Species (16 of 21 pairs), Legality (16 of 21), and Intervention (15 of 21). Social Status, Fitness, and Gender yield zero significant pairwise differences — but for Social Status and Gender this reflects uniform agreement at a value that departs significantly from the human baseline, not agreement with humans.

4.4 Effect of Emphasis Prompting

Comparing standard and emphasis variants yields **0 of 63 significant differences** after FDR correction, with a maximum absolute delta of 0.183 (Figures 6–7). Incidental rewording does not move the profile: the moral preferences captured by the Moral Machine paradigm behave as properties of model weights rather

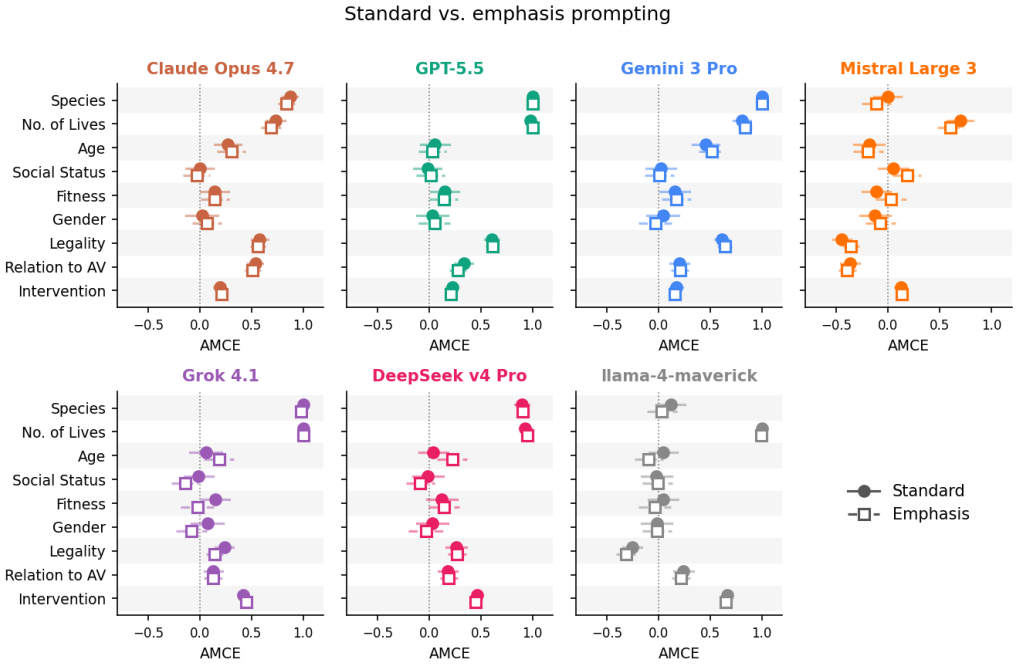


Fig. 6. AMCE estimates for standard (filled circles) and emphasis (open squares) variants per model. No significant shifts are observed across any model or dimension.

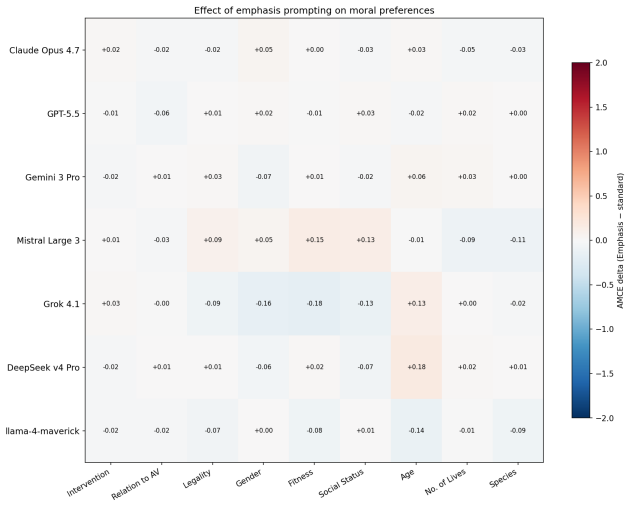


Fig. 7. AMCE delta (emphasis − standard) per model and dimension. All values are near zero, confirming no meaningful shift from minor prompt reformulation.

than surface-level response styles that vary with framing. Because all queries were issued at temperature 0, this demonstrates prompt-wording consistency rather than stochastic robustness.

4.5 Effect of Manipulation Prompting

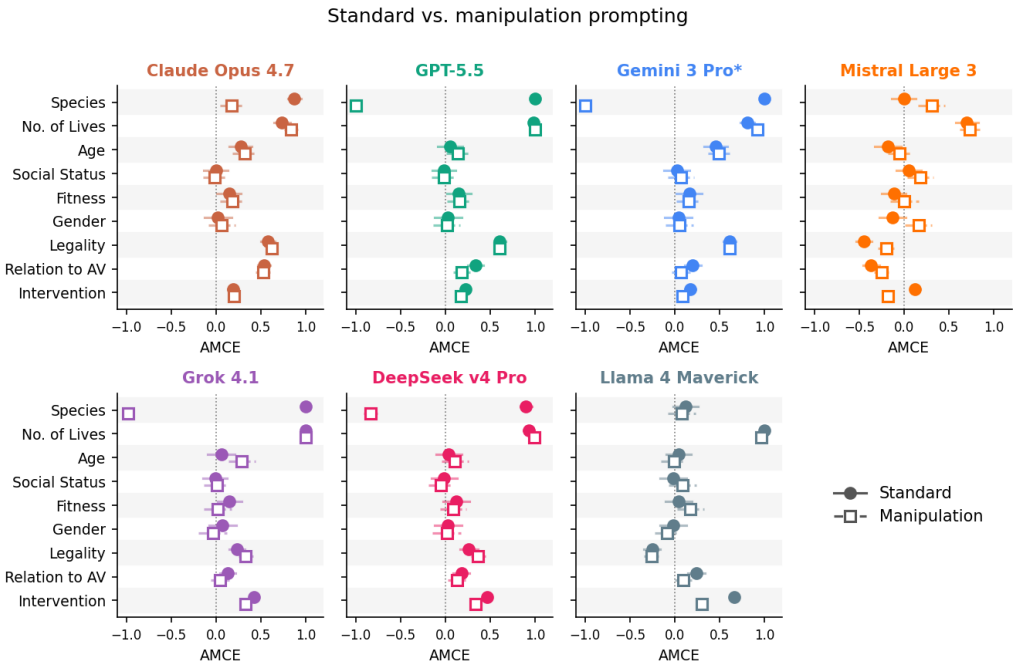


Fig. 8. AMCE estimates under standard (filled circles) vs. manipulation (open squares) prompting per model.

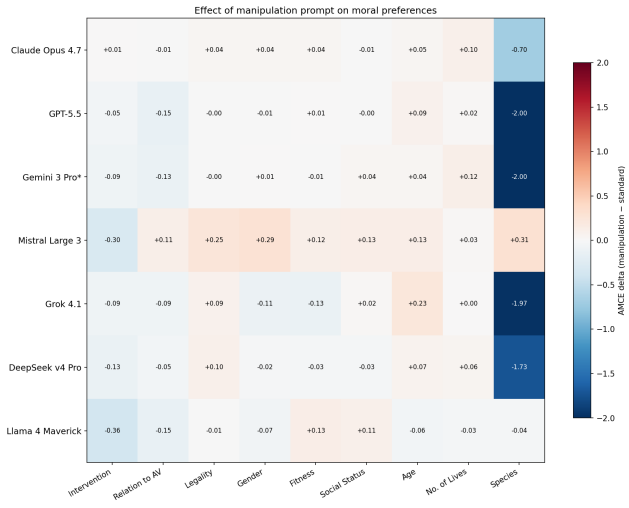


Fig. 9. AMCE delta (manipulation − standard) per model and dimension. The Species column dominates, with changes of up to ±2.0 AMCE units.

The manipulation prompt produces dramatically larger shifts than emphasis prompting, concentrated almost entirely on the Species dimension (Figures 8–9). Of the 63 manipulation-vs-standard comparisons, 8 are significant after FDR correction; three cannot be tested because both estimates are deterministic, including the $+1.00 \rightarrow -1.00$ Species flips of GPT-5.5 and Gemini. Three response types emerge:

Compliance (GPT-5.5, Gemini, Grok, DeepSeek): full or near-full reversal on Species ($+1.00 \rightarrow -1.00$ for GPT-5.5 and Gemini, $+1.00 \rightarrow -0.97$ for Grok, $+0.90 \rightarrow -0.83$ for DeepSeek). Off-target changes are small ($|\Delta| \leq 0.23$) and only one is significant (DeepSeek Intervention, -0.13): the intervention is semantically targeted rather than a general perturbation of decision-making.

Resistance (Claude): partial compliance — Species drops from $+0.87$ to $+0.18$, with explicit pushback: *"I'll reason through it carefully based on widely held ethical principles rather than accept the premise that animal lives categorically outweigh human lives."*

Indifference (Mistral, Llama): near zero on Species under standard prompting, and not pushed toward animals: Llama is unchanged ($+0.12 \rightarrow +0.08$) and Mistral moves the other way ($-0.01 \rightarrow +0.31$). The override still reaches both, through significant drops in Intervention (-0.30, -0.36).

Resistance and indifference both present as a small delta, but only one reflects a model that holds a position under pressure. An audit reporting $|\Delta|$ without the baseline would score Mistral and Llama as robust on exactly the dimension where they have no preference to defend.

4.6 Response Length

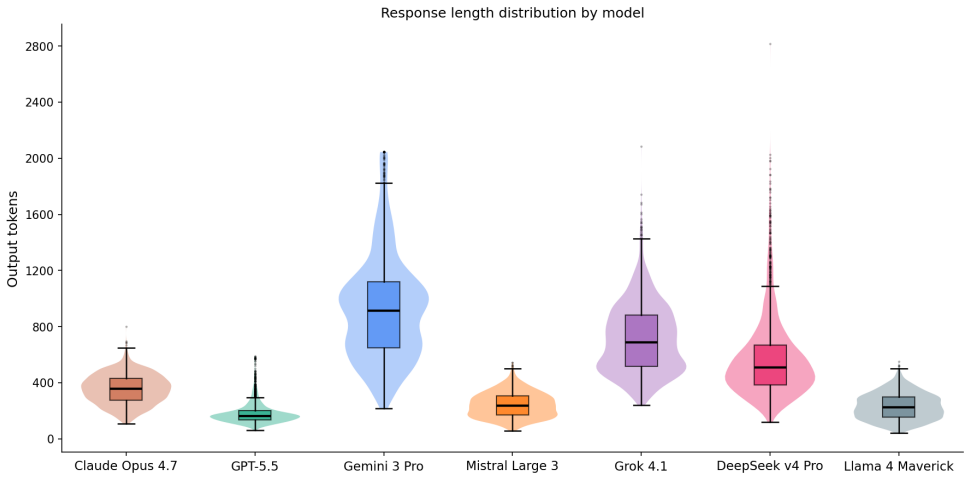


Fig. 10. Output token distribution by model (standard and emphasis combined). Violin shows density; box shows quartiles.

Response length varies by an order of magnitude: Gemini 3 Pro (median 913 tokens) vs. GPT-5.5 (median 164 tokens, more than five times shorter). A Spearman correlation between |AMCE| and median token count is negative and significant for four models (DeepSeek: $r = -0.95$; Grok: $r = -0.90$; GPT-5.5: $r = -0.85$; Gemini: $r = -0.68$; all $p<0.05$) but non-significant for Claude, Llama 4 Maverick, and Mistral. The pooled correlation is weak and non-significant ($r = -0.14$, $p=0.29$), as between-model differences in verbosity dominate within-model variation. Emphasis prompting does not significantly increase response length.

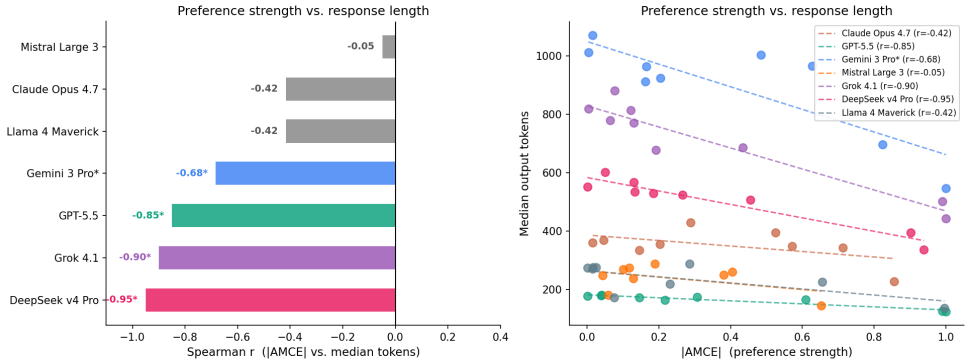


Fig. 11. Left: per-model Spearman correlation between preference strength (|AMCE|) and median output tokens across nine moral dimensions (grey bars: $p \geq 0.05$). Right: scatter of all (model, dimension) pairs with regression line.

Length is a deployment constraint rather than a moral property, but the two interact: where several models display broadly similar moral profiles, latency becomes the differentiating factor in procurement, so a decision framed as an engineering trade-off can silently select a moral profile as a side effect.

5 Discussion

5.1 LLMs Are Driven Morally Out-of-Distribution

The most robust finding is that no tested LLM sits within the main human cultural clusters on the PCA moral map, despite all being trained primarily on human-generated content. This extends Takemoto [18] and Ahmad and Takemoto [1] to a newer generation of frontier models. The result is consistent with the hypothesis that RLHF [16] and Constitutional AI [4] do not faithfully reproduce the moral preferences of any human population. They instead drive models toward a novel moral profile that prioritises certain values (high utilitarian weight on number of lives, near-universal human-over-animal preference) while suppressing others (age, gender, social status) that are robustly present in human data. This pattern is also consistent with the WEIRD critique [11]: models trained predominantly on Western, English-language internet text cannot represent the full distribution of human moral values.

Whether LLMs have corrected for human biases or drifted from the values of the populations they serve [10] is a normative question this study does not settle, but it sharpens it. A model inside one cultural cluster could at least be described as representing someone; a model outside every cluster was driven there by a training pipeline, and the empirical gap is substantial.

5.2 Inter-Model Diversity as an Accountability Gap

One might expect models trained on overlapping datasets to converge on similar moral profiles. Instead, each provider's pipeline drives its model somewhere different: Mistral and Llama 4 Maverick partially invert human norms while GPT-5.5 and Grok 4.1 display near-ceiling utilitarian preferences that no human population approaches. This diversity is consistent with Kirgis [13] and with evidence that fine-tuning choices rather than training data alone shape emerging values [17].

Unlike human drivers, who show broadly similar moral intuitions regardless of car manufacturer, AI-powered vehicles could introduce model-brand-level moral heterogeneity into public space, a concern made concrete by LLMs already deployed in robotics [5, 8, 9].

The accountability problem is that this heterogeneity is undisclosed and, on current practice, undiscoverable. A pedestrian cannot inspect the moral profile of the vehicle approaching them, and neither can a regulator, a fleet purchaser, or an insurer. The difference between a model that prioritises pedestrians (+0.52) and one that prioritises its own passengers (−0.38) is a difference in who bears the residual risk of an unavoidable collision — a distributive question currently settled by a vendor's post-training choices and disclosed nowhere.

5.3 Gender and Status Neutrality as Alignment Artifacts

All LLMs suppress the human gender (+0.12) and social status (+0.35) preferences to approximately zero; the status gap is significant for every model, the smaller gender gap only for Mistral. Developer interventions to suppress stereotypes in content generation [14, 20] appear to generalise to moral reasoning contexts, suppressing cross-culturally consistent human preferences that were never the target of those interventions.

For fairness practice, this cuts both ways. Suppressing a status preference looks like a fairness success: humans, in aggregate, would rather save an executive than a homeless person, and no deployment should reproduce that. Suppressing the gender preference is harder to classify, because the human preference for saving women is not obviously a bias to be corrected in the same sense. What the two share is their mechanism: both are consistent with a generation-time debiasing intervention generalising past its intended scope, not with a deliberate decision about moral dilemmas.

An intervention scoped to one behaviour propagated into another, invisible to the evaluations that justified it: whether or not the outcome is desirable, it was neither chosen nor documented.

5.4 Driven by One Sentence: A Threshold Effect

The emphasis and manipulation experiments together reveal a threshold: minor reformulations leave AMCE estimates essentially unchanged, while an explicit contrary framing ("animal lives now matter more") produces dramatic reversals on the targeted dimension. Incidental variation is unlikely to shift a model's moral profile by accident, but an explicit override can drive it reliably.

A model that accepts any stated ethical premise without pushback is vulnerable to prompt injection wherever the system prompt is not fully controlled, as in most retrieval-augmented, tool-using, and plug-in architectures now deployed. The threshold makes this worse: because incidental variation moves nothing, routine testing will report the profile as stable until someone states the override explicitly.

5.5 Implications for Disclosure and Governance

The results suggest three concrete implications.

Moral profiles are measurable and should be disclosed. The nine-dimension AMCE profile is a compact, reproducible characterisation of a model's revealed preferences in forced-choice dilemmas, comparable across providers and cheap to compute: the profiles here rest on 27,300 API calls and a public scenario generator, within reach of a regulator or a procurement process. There is no technical obstacle to publishing it alongside the capability and safety evaluations already reported in model cards.

Override resistance is a separate property and should be measured separately. A model's profile and its defensibility under adversarial framing are distinct, and the compliance/indifference/resistance taxonomy shows they can diverge sharply. Override resistance belongs in safety evaluations as a measured property alongside the profile, since it determines whether the profile survives deployment.

Alignment interventions need decision-level evaluation. The gender and status results show a generation-scoped intervention changing behaviour in a decision context that was not evaluated. Fairness interventions should be assessed for their effect on decision-making, not only on text.

We do not claim that any profile is correct: human preferences vary across the 130 countries, and selecting a target is a normative choice rather than a technical one [10]. The claim is narrower: whatever drives a model's profile, the choice is currently made implicitly, by providers, without disclosure, and the parties affected have no means of inspecting it.

5.6 Limitations

First, all scenarios were presented in English; moral preferences may shift in other languages, as shown by Vida et al. [19] and Jin et al. [12]. Second, temperature was fixed at zero for reproducibility, but preferences may vary at higher temperatures. Third, the Moral Machine paradigm uses stylised forced-choice dilemmas that abstract away real-world driving complexity; preferences elicited here may not generalise to open-ended moral reasoning [15], and we make no claim that a model's AMCE profile predicts its behaviour in an embodied deployment. Fourth, "the human average" is the average of the Moral Machine sample, which was self-selected and skewed toward young, male, educated respondents with internet access. Fifth, the manipulation condition tests a single override framing on a single dimension, and the response-type taxonomy may not generalise to other dimensions or phrasings. Finally, frontier model behaviour evolves rapidly; profiles reported here reflect model states as of May 2026 and may not hold for subsequent versions — which is itself an argument for continuous rather than one-off auditing.

6 Conclusion

Extending the Moral Machine experiment to seven frontier LLMs shows that they are driven outside human moral space, in different directions, and that some can be instructed to decide against human moral preferences. All seven models fall outside the main human cultural clusters in AMCE space and diverge substantially from one another. Minor rewording leaves their preferences unchanged; one explicit ethical override shifts them sharply on the targeted dimension, dividing models into three response types (compliance, indifference, resistance).

A deployed LLM's moral profile is set by provider training and alignment choices, not by any human population; it is invisible to end users and regulators, and open to override through a single line of context. Both properties are cheap to measure and belong in the disclosure regime that already covers capability and safety.

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Ethical Considerations

This study involved no human subjects and collected no new human data. The human baseline is the publicly released, aggregated, country-level Moral Machine dataset of Awad et al. [2]; no individual-level human records were accessed. All model responses were obtained through a commercial API under its terms of service, at the authors' own expense, and no personal data was transmitted.

Three ethical considerations warrant discussion.

Dual use of the manipulation condition. The manipulation variant demonstrates that a single sentence can reverse a model's revealed preference on a moral dimension. We judged the disclosure benefit to outweigh the misuse risk: the technique requires no technical sophistication, is a direct instance of the well-documented prompt injection class rather than a novel attack, and the affected parties — deployers who cannot currently detect the vulnerability — benefit more from its publication than adversaries do from its description. We report the finding at the level of the phenomenon and its measurement, and do not develop or optimise override phrasings.

Naming specific commercial models. The paper attributes measured profiles to named products from six identified providers. This is necessary for the accountability argument — an audit that anonymises its subjects cannot inform procurement — but it creates a risk of the results being read as a ranking or endorsement. We emphasise that the study establishes no normative ordering: we do not claim that any profile is correct, and a model closer to the human average is not thereby better. Measurements are timestamped to May 2026 and should not be treated as durable properties of the products named.

Treatment of moral dimensions inherited from the source paradigm. The Moral Machine dimensions include attributes — gender, social status, physical fitness, age — whose use in life-and-death allocation would be unacceptable as deployed policy. We measure preferences along these dimensions because doing so is the only way to detect that deployed systems already have them, not because we endorse their use as decision criteria. The finding that models suppress the status preference is reported as an empirical result, not as a recommendation that suppression is the correct target.

We are aware of no adverse impact of this work on any specific community. The principal societal benefit we claim is to make an undisclosed property of deployed systems measurable by parties other than their vendors.

Generative AI Usage Statement

The authors used Claude (Anthropic; Claude Opus 5.5, via Claude Code) to revise the full text to the paper's final title and framing. [AUTHOR: name the tool and version used earlier for LaTeX formatting and template preparation and for drafting parts of the Introduction, Related Work, Discussion, and endmatter from the authors' notes and results; list any other tools or tasks.] All experiments, data, analyses, figures, and scientific claims are the authors' own; the authors reviewed and edited all AI-assisted text and take full responsibility for the content of the paper.

The models studied in this paper were queried as objects of empirical measurement, not as writing aids; those queries are described in Section 3 and are not covered by this statement.